

Bayesian Modeling of Cyclone Intensity

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1 Introduction

Cyclones, consisting of hurricanes, typhoons, and other tropical storms caused by systems of low atmospheric pressure, are a yearly hazard to global citizens and the regions in which they live. The ability to detect cyclones early-on and predict their development is crucial to protecting human life and property.

Our project aims to use the SHIPS cyclone dataset and a Bayesian hierarchical model to predict the maximum sustained wind speed 36 hours after a cyclone is first detected. The maximum sustained wind speed is one way to classify the severity of a cyclone, thus providing a way for a model like ours to be used to assist in emergency-preparedness and disaster-relief efforts.

In this report, we determine a relationship between five predictive factors and maximum wind speed, and test whether the relationship between the maximum wind speed and these factors is dependent on the ocean basin in which the cyclone occurs.

2 Data

The Statistical Hurricane Intensity Prediction Scheme (SHIPS) is a model produced by the National Hurricane Center (NHC) to forecast the progression of tropical storms in the Atlantic, eastern North Pacific, and central North Pacific basins [RAM22]. The dataset used in the development of this model is publicly available. We chose to use this dataset due to its reputability and inclusion of certain helpful features. It contains data for 749 eastern North Pacific, 610 Atlantic, and 57 central North Pacific storms. After filtering by storms that continued for at least 36 hours and had no missing measurements in the features we wished to include, there were 555 eastern North Pacific, 548 Atlantic, and 23 central North Pacific storms remaining.

For the duration of each storm, the dataset reports measurements of climate variables like maximum sustained wind speed (VMAX) and minimum sea pressure (MSLP) at 6 hour intervals. The dataset also includes predictors that have been calculated from atmospheric data. The only such predictor we use in our model is Maximum Potential Intensity (VMPI), which is a function of sea surface temperature (SST) and atmospheric outflow temperature calculated in the method given by Bister and Emmanuel (2002) [BE02]. From MPI we calculated Intensity Potential (POT) by subtracting the storm's intensity at 12 hours from its MPI. This feature has been shown to have predictive value [QL18]. We also calculated an approximate intensity gradient from hours 6 to 12 (VGRAD6-12) by subtracting the maximum sustained wind speed at 6 hours from the maximum sustained wind speed at 12 hours and dividing by 6.

We chose to join the SHIPS dataset with Oceanic Niño Index (ONI) data from the National Weather Service. This index is a three month running average of SST anomalies in the Niño 3.4

region [NOA]. ONI will be interesting to include as a predictor because it is understood to be a positive predictor for cyclone intensity in the central and eastern Pacific basins and a negative predictor for cyclone intensity in the Atlantic basins [Bel14]. The SHIPS dataset reports only the month and year of each storm, so we will predict based on the ONI on the first day of the month of occurrence.

3 Methodology

3.1 Model Variables

We wish to calculate the maximum sustained wind speed from 12 to 36 hours after first measurement. Our predictors are as follows:

1. ONI: the El Niño Index on the first day of the month in which the cyclone is recorded
2. logVMAX12: the log of maximum sustained wind speed from 0 to 12 hours after first measurement
3. MSLP12: the minimum Sea Level Pressure over the same time period as logVMAX12
4. POT12: the maximum potential intensity over the same time period as logVMAX12
5. VGRAD6-12: the gradient in the maximum sustained wind speed between 6 hours and 12 hours after first measurement
6. BASIN_IDX: a categorical variable marking which ocean basin which the storm occurs in

We assume that the basin which the storm develops in will control the parameters governing the model, rather than being a separate predictor.

3.2 Bayesian Hierarchical Model

To ensure that the maximum sustained wind speed (W_i) is a positive value, we specify the marginal distribution on the log of the wind speed as:

$$\log W_i | \mu_i \sigma_{b_i}^2, \sim N(\mu_i, \sigma_{b_i}^2)$$

Note that in our model specification, the subscript b refers to the set of parameters for a basin. Thus, σ_b^2 is the variance of the maximum sustained wind speed for the b -th basin.

We take μ_i to be:

$$\vec{x}_i \vec{\beta}_{b_i}^T$$

Our priors for our latent parameters are as follows:

$$\sigma_b^2 | \alpha, \gamma \sim \text{Inverse-Gamma}(\alpha, \text{scale} = \gamma)$$

$$\vec{\beta}^{(d)} | \nu_d, \tau_d \sim \text{MultivarNormal}(\nu_d \vec{1}, \tau_d^2 I_B)$$

$$\tau_d^2 | \lambda_d \sim \text{Inverse-Gamma}(a, \text{scale} = \lambda_d)$$

We use inverse-gamma distributions on σ^2 and τ^2 to ensure that these values are positive.

$$\alpha \sim \text{Gamma}(y_\alpha, \text{scale} = z_\alpha)$$

$$\gamma \sim \text{Gamma}(y_\gamma, \text{scale} = z_\gamma)$$

$$\nu_d \sim \text{MultivarNormal}(\vec{0}, v^2 I_D)$$

$$\lambda_d \sim \text{Gamma}(y_\lambda, \text{scale} = z_\lambda)$$

Note that the subscript d refers to the set of predictor variables (5 in our case).

Here $a, y_\alpha, z_\alpha, y_\gamma, z_\gamma, y_\lambda, z_\lambda$, and v are all constant values.

3.3 Justification of Prior Distributions and Parameters

When choosing our likelihood distribution, we considered that wind speed is always non-negative. We decided to predict on the log of wind speeds in order to preserve positivity, and after plotting histograms of wind speeds from our data, we observed that the log of these values appeared roughly normally-distributed. This makes sense because we expect low-wind-speed storms to occur much more frequently than high-wind-speed storms, and taking the log of these values “evens out” the spread to appear more bell-shaped. After determining the likelihood, we chose the rest of our prior distributions by considering which parameters should be constrained to be non-negative and prioritizing conditional conjugacy within our model for ease of implementation. However, we also considered our model with alternate priors on τ_d and λ_d , defined by $\tau_d | \lambda_d \sim \text{Cauchy}^+(\lambda_d)$, $\lambda_d \sim \text{Cauchy}^+(m)$. These alternative priors are discussed further in Section 4.4.

In eliciting prior parameters, we used prior understanding as well as general information found online to estimate a mean for each desired distribution. Because we did not have strong confidence in our estimates, we adjusted the shape parameters of these prior to be weakly informative. A summary of our parameter selection process is given below. In order to make the priors weakly informative, we used values between 1 and 3 for the shape parameters of Gamma and Inverse-Gamma distributions without further justification.

- Because we didn’t expect a linear model to fit the data extremely well, we wanted to allow reasonably large deviation – we chose 15 knots, or roughly 2 in log-knots – from the regression line. Therefore, we wanted the prior on σ_b^2 to have a mean of 2. To do this, we set $\alpha = 3$, which necessitates $\gamma = 8$. This gives target means for our hyperpriors on α and γ .
- To allow the prior on α to have a mean of 3, we chose $y_\alpha = 2, z_\alpha = \frac{3}{2}$.
- To allow the prior on γ to have a mean of 8, we chose $y_\gamma = 2, z_\gamma = 4$.
- Because log wind speed is a predictor, we expected the output of the model to be on roughly the same scale as the predictors. This means that the regression parameters, whose means are given by $\vec{\nu}$ should not be too large. We set $v^2 = 100$, meaning that we expect the regression parameters to mostly fall within $[-30, 30]$.
- We expected that the regression parameters for a single predictor across all basins will be within ± 15 of each other, so we thought that the mean of the prior on τ^2 should be around 25. We set $a = 3$, and this means that the mean of the prior on λ_d should be 50. Choosing $y_\lambda = 2$ necessitates $z_\lambda = 25$.

3.4 Model Fitting Algorithm

The full conditionals for each parameter are conjugate, except for the full conditional on α . Thus, our MCMC algorithm uses Gibbs samplers for all parameters except for α , which uses a Metropolis-Hastings sampler.

In comparison to a STAN sampler, we found that our sampler ran roughly 10x faster, requiring on average 0.0037 seconds per iteration compared to STAN’s average of 0.0314 seconds. Our

sampler and STAN’s sampler converged to similar values for all parameters (See Figure 1), and gave similar results in terms of convergence diagnostics (ESS and $\hat{R}$). Overall, leveraging the conjugacy of our model allowed our sampler to produce good results in terms of posterior draws at a relatively low computational cost.

4 Results

4.1 Main Analysis and Convergence Diagnostics

We sampled multiple chains on both our own MCMC sampler and a STAN implementation, and observed that long chains converged to the same parameter values, with a small margin for Monte Carlo error (see Figure 1). Due to the large number of parameters in our model (35 total), we needed to run fairly long chains to reach desired convergence diagnostics. We found that running 2 chains with 30,000 iterations each and a burn-in period between 500 and 1,000 iterations produced posterior draws with a minimum ESS of roughly 3,500 and a maximum $\hat{R}$ of 1.001 for all parameters. All of the trace plots looked reasonable (posterior draws oscillated around mean values and did not get stuck in any locations), the acceptance rate for the Metropolis-Hastings sampler for α varied between 0.4 and 0.65 across different chains, and running longer chains did not produce divergence from our results. All of these diagnostics together present strong evidence for convergence.

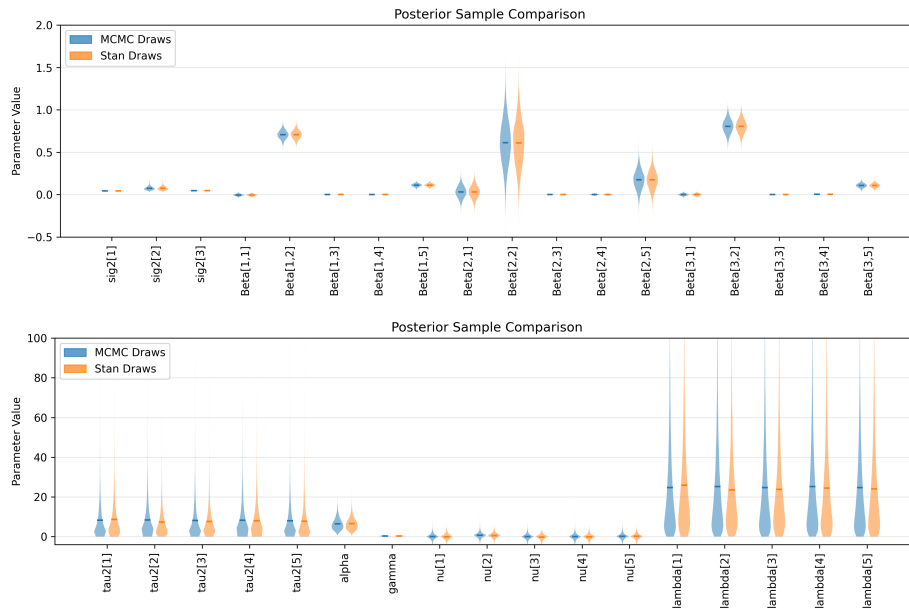


Figure 1: Comparison between the retrieved parameters of our Python implementation of the MCMC sampler and the STAN implementation. Retrieved parameter values and distributions are largely identical.

The most interesting result of our main analysis is the convergence of the regression parameters, as many of them were shrunk to nearly zero. Out of our 5 predictors, the model seemed to rely almost exclusively on the log of the current wind speed ($\beta^{(2)}$) and the wind speed gradient over the most recent 6 hours ($\beta^{(5)}$). Because we assumed linearity, it makes sense that the the model for future wind speed would be almost entirely determined by current wind speed and an estimation of the slope. This reveals one weakness of our model, namely, the linear assumption

and lack of physical dynamics involved in our model limits the ability to capture complex/non-linear relationships in predictors such as the minimum sea pressure. Furthermore, we observed that the regression parameters turned out to be roughly equal across basins, meaning that the linear relationship between wind speed and our predictors does not vary significantly in different ocean basins.

4.2 Posterior Predictive Checks

To check that our model fit the data, we produced quantile-quantile plots for posterior predictive draws of max wind speed for every storm in the dataset. We found that the points show only minor deviations from the uniform CDF, which leads us to believe that this model is a good fit for the data. We also used posterior predictive samples to model the probabilities that a storm would reach or exceed its realized storm category. We chose to calculate the diagnostic this way because in the context of deciding what action to take to prepare for a storm, underestimating intensity is more dangerous than overestimating. We found that while this probability was usually above 80%, for more severe storms it tended to be lower. Overall, Our model tends to underestimate the wind speed for more severe storms. This could be because our dataset is heavily weighted towards smaller storms, which occur much more often.

We chose to visualize our results by creating histograms of posterior predictive draws using storms from our dataset so that we have both the set of features to predict on and the true value to compare to. The Saffir-Simpson scale wind speed thresholds are represented by the colored vertical bands (See Figure 2).

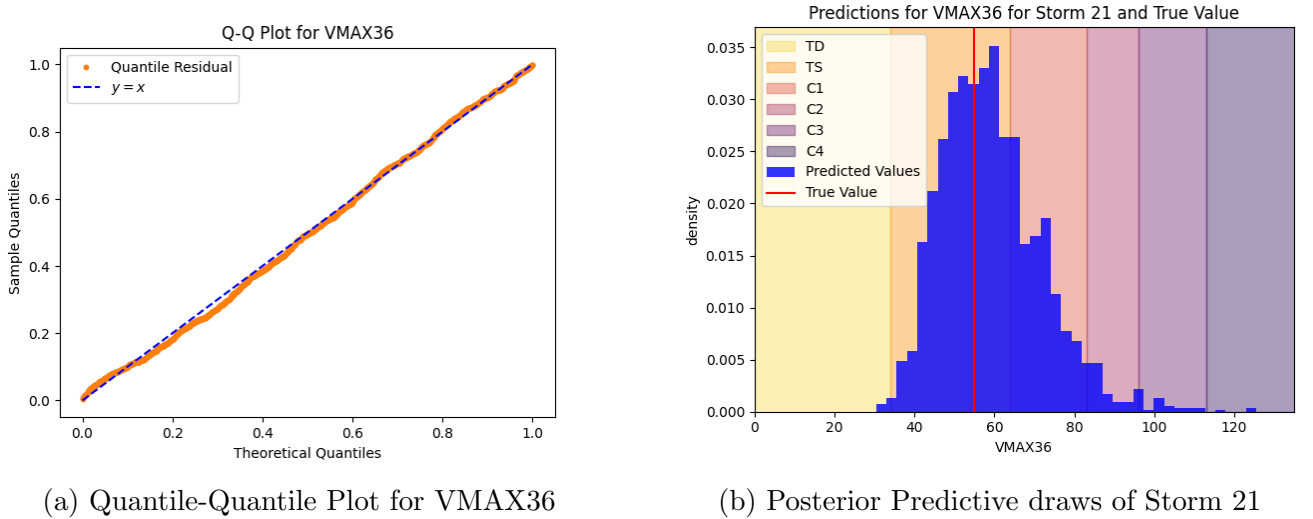


Figure 2

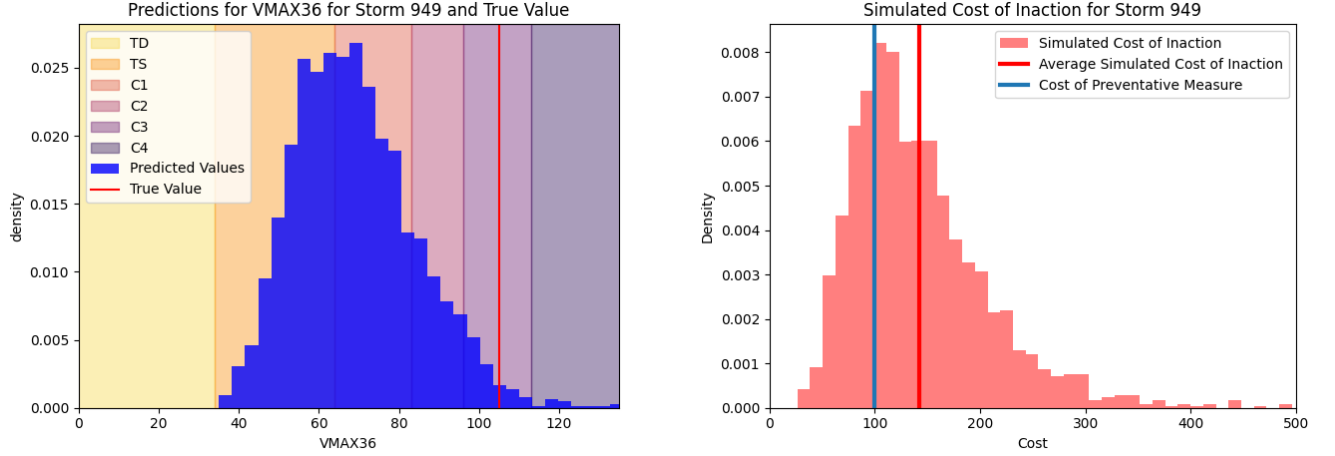
4.3 Decision Analysis

The primary advantage of a Bayesian model for this task is its ability to represent uncertainty, which would allow the NHC or other agencies to make good decisions about what recommendations to make. Predictions from our model could be seamlessly connected to a given cost function, where the costs associated with the inconvenience of unnecessary storm preparations could be encoded alongside the costs associated with the dangers of making insufficient preparations.

Decision analysis could be performed using Saffir-Simpson categories or a continuous cost function. For example, if A is a set of possible actions and $C(a, \theta)$ is the cost of action a given

the maximum sustained wind speed reaches θ in the next 24 hours, and s_i is the i th Monte Carlo prediction of VMAX36, the ideal action would be estimated by the following formula: $a^* = \arg \min_{a \in A} \frac{1}{N} \sum_{i=1}^N C(a, s_i)$. We provide an example of the decision analysis process. The numbers used in our cost function may not be entirely realistic, but we choose for the cost of inaction to exceed the cost of preventative measures beginning at a C2 hurricane.

Suppose that implementing a certain preventative measure a_p has a fixed cost, represented by $C(a_p, \theta) = 100$, and that the cost of inaction a_n increases quadratically with future storm intensity, represented by $C(a_n, \theta) = x^2/50$. We show the results of decision analysis for one storm observation in Figure 3.



(a) Posterior Predictive Draws for Storm 949

(b) Draws for cost of Inaction for Storm 949

Figure 3: Decision analysis for Storm 949. Posterior predictive draws show that our model drastically underestimates the storm’s severity. However, simulated draws of the cost of inaction show that the average simulated cost of inaction is greater than the cost of preventative measures. This indicates that the optimal choice is to employ preventative measures.

4.4 Prior Dependence Analysis and Model Criticism

To test the fit of our priors, we swapped the inverse-gamma prior on τ with a Half-Cauchy prior, with the same hyper parameters now drawn from a half-Cauchy distribution and trained on the data. This switch in prior distribution corresponds to an assumption that the regression parameters may be the same across all basins, since the Half-Cauchy distribution allows the between-group variance to be shrunk to zero. After sampling from both models, we found that the regression parameters (σ_b^2 and $\beta_b^{(d)}$) converged to the same values (other parameters did not converge to the same values, but this is due to the fact that they have different interpretations under each model). We compared the WAIC and LOO-IC for both model, as displayed in Table 1.

The results suggest that the half-Cauchy prior is marginally better than the Inverse-Gamma prior, both in terms of the WAIC and LOO-IC values and the number of parameters. We can interpret the smaller number of effective parameters from the Half-Cauchy model as a more parsimonious model, which is to be preferred. Furthermore, the retrieved parameters between the two models are broadly consistent with each other, though the Half-Cauchy model produces smaller posterior sample variances for most parameters. The fact that the Half-Cauchy model outperformed the Inverse-Gamma model suggests that the underlying effects of each predictor on max wind speed do not vary significantly across different ocean basins.

Model	WAIC	LOO-IC	P-WAIC	P-LOO
Inverse-Gamma	-258.82	-258.02	16.77	17.17
Half-Cauchy	-268.17	-267.98	12.24	12.34

Table 1: WAIC and LOO-IC calculations for the two prior cases. Values have been rounded to the nearest hundredth.

If we were to extend this analysis, we would consider replacing the distributions on both τ and σ with half-Cauchy distributions. This would require replacing two Gibbs samplers within our MCMC algorithm with Metropolis-Hastings samplers, which may increase the number of iterations needed for convergence.

4.5 Comparison with Non-Bayesian Analysis

Our first non-Bayesian approach to cyclone intensity prediction was a classical linear regression on max wind speed using the same predictors as in our Bayesian modeling. This is less flexible than our hierarchical model, because it does not allow for features to have different coefficients depending on the basin. The next approach was to separate the data by basin, then perform a linear regression on storm intensity for each basin independently. This allows for the flexibility of different coefficients for each basin, but it also shrinks the dataset drastically, especially for the central North Pacific (CP) basin. This model assigned positive weight to ONI in the CP basin, as well as significantly different intercepts and weights of logVMAX12.

We also trained a Random Forest Regressor (RFR), which is theoretically able to capture non-linear correlations between the features and the target. The RFR assigned low but nonzero importances to the basin indicators, with a combined importance of 4.8% for storm basin. This model also gave more weight to ONI and POT12 than the previous linear models, which indicates there may be nonlinear correlations between these features and the target. Finally, to investigate whether storm intensity has a continuous spatial dependence, we fit another RFR and removed the basin indicator variable, opting for exact storm latitude and longitude (at $t = 0$) from the SHIPS dataset. This model assigned weight to both latitude and longitude, with a combined importance of 18.5% for both spatial variables.

In order to compare these results with our Bayesian model, we obtained point-estimates by taking the mean of posterior predictive draws for each observation. In terms of MSE and explained variance, the Bayesian out-performed both linear models, and was only slightly worse than the RFR, as shown in Table 2.

Model	MSE	Explained Variance
Bayesian Linear Regression	0.0452	0.426
Linear Regression (combined basins)	0.0462	0.420
Linear Regression (avg. over separate basins)	0.0539	0.336
Random Forest Regressor (discrete basins)	0.0360	0.563
Random Forest Regressor (continuous lat/long)	0.0373	0.514

Table 2: Model performance comparison in terms of MSE and explained variance between Bayesian linear regression and various non-Bayesian regression models.

5 Conclusion

In this report, we developed a Bayesian hierarchical linear regression model for the maximum sustained wind speed over hours 12-36 of a tropical storm. We produced posterior parameter samples from our model using both our own MCMC implementation in Python and a STAN model, and found that both methods agree on an linear function of current maximum sustained wind speed and wind speed gradient over the past 6 hours, with current maximum sustained wind speed being the strongest predictor. We found that there is less individuality across ocean basins than we hypothesized, which is evidenced by the closeness of regression parameters across all basins and the fact that a Half-Cauchy prior on predictor variance provided a better model fit than an Inverse-Gamma prior.

On average, our model predicted with 80% certainty that a storm would reach or exceed its actual attained severity category, which is a promising result for decision analysis. However, we acknowledge that our current model tends to underestimate the most severe storms. This is likely a result of our model lacking physical dynamics to capture complex relationships between predictors and the target, as well as insufficient sample size for severe cyclones. So, while our current results are meaningful and indicate valuable directions for future research, it would be irresponsible to make any real decisions based on them.

We suggest two possible extensions for our model: first, the incorporation of non-linear interactions between ONI, MSLP12, and POT12 to capture the effects of these parameters on storm development. Second, we suggest that our training dataset be extended to include more equal numbers of storms of all intensity categories (perhaps by augmenting the dataset with numerical simulations). Either of these two extensions would make our model more usable for decision-making when it comes to reducing the harm posed by cyclones.

References

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A Code

All code used in the making of this report can be found in the Cyclones GitHub repository located at <https://github.com/quinn-oveson/cyclones.git>.

A list of important code files, along with brief descriptions, is provided below:

- `sampler.py`: contains fully-developed versions of the `CycloneSampler` class (our MCMC implementation for sampling posterior parameters) and the `CycloneDataSimulator` (generates simulated data from our specified model). Both classes can be imported from this script into other Python scripts/notebooks.
- `sampler_dev.ipynb`: contains specification of our model and our work for implementing and testing the `CycloneSampler` and `CycloneDataSimulator` classes.
- `stan_dev.ipynb` : contains our testing for the STAN implementation of our solver, which can be found in `conjugate_model.stan`.
- `sensitivity_analysis.ipynb` : contains our testing of our sampling model's sensitivity to number of observations and prior parameters, as described in Appendix C.
- `prior_dependence_analysis.ipynb` : contains our tests for the dependence of our results on the retrieved parameters, as described in Section 4.4
- `convergence_comparison.ipynb` : contains our testing and checks to show that our MCMC sampler converges to the same posterior parameters as a STAN sampler.

B Disclosure of AI Usage

Zac acknowledges the use of AI to provide needed syntax and libraries for handling the SHIPS dataset and LaTeX formatting.

Tiara acknowledges the use of AI in assisting with the following tasks:

- Brainstorming – described some of the project criteria as well as our group's topics of interest and asked for dataset recommendations
- Derivations – asked didactic questions/general examples of deriving posterior distributions and implementing hybrid samplers. No specific information about our model or code was provided to AI.
- Coding help – looked up documentation and syntax for plotting functions, Stan methods, and LaTeX. Used AI to assist in debugging strange errors and catching typos in very large code blocks. No code was generated by or copy-pasted from AI, with the exception of some plotting code.

Quinn acknowledges using AI for the following purposes:

- Asked for details of pandas join and concatenate functions
- Got help with installations/dependencies issue
- Used to look for sources that describe the effect of El Nino Index
- Prompted to make side by side pie chart graphing function that dealt with spacing issues

C Sensitivity Analysis on Simulated Data

After implementing our own MCMC sampler for our model, we tested its convergence by simulating data from our model and assessing our sampler’s ability to recover the true parameters. Plotting 95% credible regions of our posterior draws showed adequate coverage of the true value across all parameters, as shown in Figure 4.

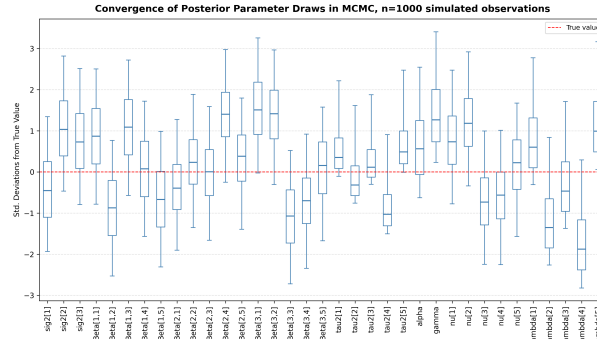


Figure 4: 95% credible regions for posterior parameters draws, scaled in number of posterior standard deviations away from the true parameter value.

We also examined the sensitivity of our sampler to varying numbers of observations and mis-specified prior parameters. We found that although a small dataset ($N = 100$) can accurately recover some of the parameters, a sample size of at least $N = 1000$ is needed to reliably capture all of the true values within a credible region and to shrink the posterior variance. Our dataset included $N = 1171$ cyclone observations, which we think is sufficient based on this analysis. We also found that the predictor-specific parameters (τ_d^2 and λ_d) were better-identified when data was provided from at least 5 ocean basins. This is a potential source of uncertainty in our results, since our dataset only records observations for 3 ocean basins. A plot showing the model’s sensitivity to number of observations is provided in Figure 5. The results for the model’s sensitivity to prior parameters were similar – running the sampler with mis-specified prior parameters resulted in posterior draws that usually still captured the true value, but had larger variance, offset means, and required longer chains than a sampler given correct prior parameters.

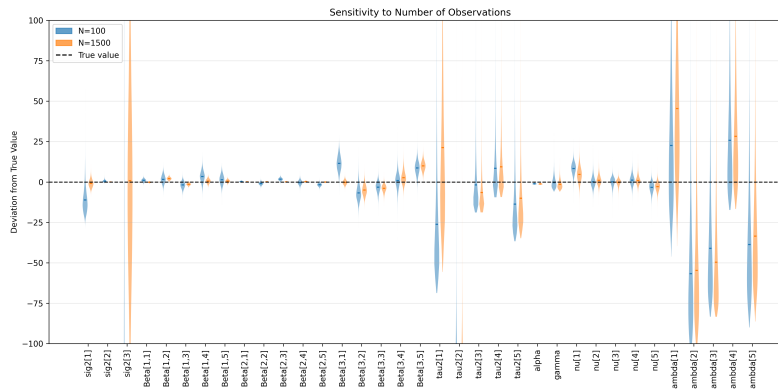


Figure 5: Distributions of posterior draws given $N = 100$ vs. $N = 1,500$ observed data points.